

Impedance & Highpass Filters

Review: Resistors, Capacitors, and Inductors

R $V=IR$ C $I=C \frac{dV}{dt}$ L $V=L \frac{dI}{dt}$

Alternating Current (AC) $\odot$

$V(t) = V_0 \cos(\omega t)$
 $I(t) = I_0 \cos(\omega t - \phi)$

Can write in terms of complex impedance:

define: $\tilde{V}(t) = V_0 e^{i\omega t} \Rightarrow e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$

$\tilde{I}(t) = \text{Re}\{\tilde{V}(t)\}$

$= C \frac{d\tilde{V}}{dt} = C \frac{d}{dt}(V_0 e^{i\omega t}) = i\omega C \cdot V_0 e^{i\omega t}$

so, $\tilde{I}(t) = i\omega C \cdot \tilde{V}(t)$

Capacitor: $\tilde{V}(t) = \tilde{I}(t) \cdot \underline{Z}_C$, where $\underline{Z}_C = \frac{1}{i\omega C}$ C impedance

Inductor: $\tilde{V}(t) = i\omega L \cdot \tilde{I}(t) = \underline{Z}_L \tilde{I}(t)$, where $\underline{Z}_L = i\omega L$ L impedance

Resistor: $\tilde{V}(t) = \tilde{I}R = \tilde{I} \underline{Z}_R$, where $\underline{Z}_R = R$ R impedance

Complex version of Ohm's Law:
Same as Ohm's for AC using $\underline{Z}$

Review of Complex #'s:

$\underline{Z} = A + iB + R e^{i\phi}$
Real Imaginary Magnitude

$R = \sqrt{A^2 + B^2}$ $A = R \cos(\phi)$ $B = R \sin(\phi)$
 $\tan(\phi) = \frac{B}{A}$ $\phi = \arctan\left(\frac{B}{A}\right)$

If same signal of $e^{i\omega t}$: Multiplying complex # by i : rotates by -90°

i.e. $-\frac{\pi}{2} e^{i(\frac{\pi}{2})} = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) = 0 + i1 = i$

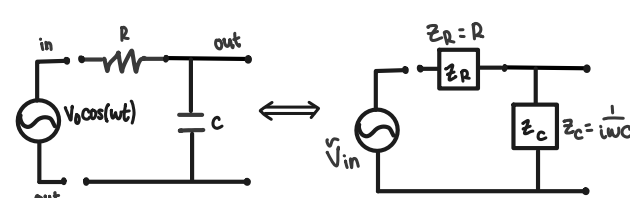
$i R e^{i\phi} = e^{i(\frac{\pi}{2})} \cdot R e^{i\phi} = R e^{i(\phi - \frac{\pi}{2})}$

Rules for Equivalent Impedance

$\underline{Z}$'s in parallel and series follow same rules as resistors
 $\rightarrow$ Loop Rule

$\tilde{V}_{AC} - \tilde{V}_1 - \tilde{V}_2 = 0$ $\tilde{V} = \tilde{I}(\underline{Z}_1 + \underline{Z}_2)$ $\tilde{V} = \tilde{I}(\underline{Z}_{eq})$
 $\underline{Z}_1$ $\underline{Z}_2$

Set up Circuit Diagram w/ Equivalent Resistances Low Pass



$V_0 \cos(\omega t) = \tilde{V}_{in} = V_0 e^{i\omega t}$

$\underline{Z}_R = R$

$\underline{Z}_C = \frac{1}{i\omega C}$

$\tilde{V}_{out} = \tilde{V}_{in} \left(\frac{\underline{Z}_C}{\underline{Z}_R + \underline{Z}_C} \right)$

$= \tilde{V}_{in} \left(\frac{\left(\frac{1}{i\omega C} \right)}{\left[R + \left(\frac{1}{i\omega C} \right) \right]} \right) \cdot i\omega C$

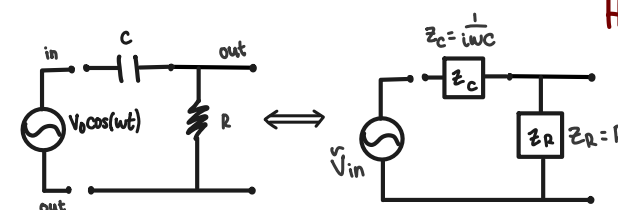
looks like a voltage divider

$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$

Gain $\left| \frac{\tilde{V}_{out}}{\tilde{V}_{in}} \right|^2 = \left| \frac{\tilde{V}_{in} \left(\frac{1}{1 + i\omega RC} \right)}{\tilde{V}_{in}} \right|^2 = \left| \frac{1}{1 + i\omega RC} \right|^2 = \left(\frac{1}{1 + i\omega RC} \right) \left(\frac{1}{1 - i\omega RC} \right)$

$\omega \ll \frac{1}{RC} \left| \frac{V_{out}}{V_{in}} \right|^2, G \rightarrow 1$
 $\omega \gg \frac{1}{RC} \left| \frac{V_{out}}{V_{in}} \right|^2, G \rightarrow 0$

Low Pass Filter



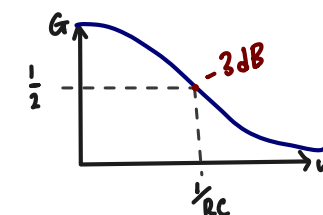
High Pass

similar steps to solve: $\tilde{V}_{out} = \tilde{V}_{in} \frac{\underline{Z}_R}{\underline{Z}_R + \underline{Z}_C} = \frac{R}{R + \frac{1}{i\omega C}} \tilde{V}_{in} = \frac{i\omega RC}{1 + i\omega RC} \tilde{V}_{in}$

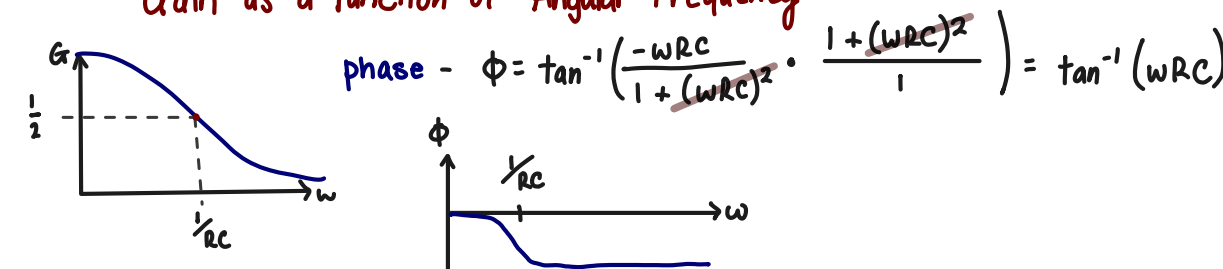
Gain, $G = \left| \frac{V_{out}}{V_{in}} \right| = \frac{(\omega RC)^2}{1 + (\omega RC)^2}$ for $\omega \ll \frac{1}{RC}, G \rightarrow 0$ $\omega \gg \frac{1}{RC}, G \rightarrow 1$ Filter

decibel scale (dB)

$dB = 10 \log_{10} \left| \frac{\tilde{V}_{out}}{\tilde{V}_{in}} \right|^2 = 2 \log_{10} \left| \frac{\tilde{V}_{out}}{\tilde{V}_{in}} \right|$
 recall, $\log(1) = 0$, $0 dB \Leftrightarrow G = 1$



Gain as a function of Angular Frequency



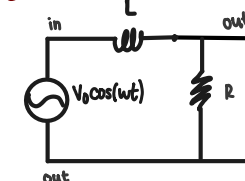
straight line on a log-log plot $\Leftrightarrow$ power law

$y = A\omega^n$

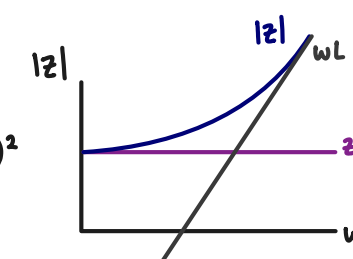
$\log y = \log A\omega^n = \log A + \log \omega^n$
 $= \log A + n \log \omega$

$f(x) = \log A + nx$

Low-Pass Filter

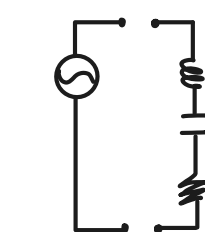


$\underline{Z} = R + i\omega L$
 $|\underline{Z}| = \sqrt{R^2 + (\omega L)^2}$



ω low frequencies, R dominates the impedance while ω high ω ($2\pi f$)
 $\rightarrow$ inductance follows the reactance of the inductor

RLC - Resonant Circuit



$\omega_0 \underline{Z}_{WL} = R$
 $V_0 = V_R = IR$

$\underline{Z}_{WL} = R + \frac{1}{i\omega C} + i\omega L$
 $= R + i\left(\omega L + \frac{1}{\omega C}\right)$

find where $|\underline{Z}_{WL}|$ is smallest (ω what ω)

$\text{Im}\{\underline{Z}_{eq}\} = \omega L - \frac{1}{\omega C} = 0$ when $\omega = \frac{1}{\sqrt{LC}}$ resonance frequency

biggest I when $\omega = \omega_0 \Rightarrow$ biggest voltage V_R

$\omega = \omega_0: \tilde{V}_C = \tilde{I} \underline{Z}_C = \left| \frac{V_0}{R} \right| \frac{1}{i\omega_0 C} = i \frac{V_0}{R} \sqrt{\frac{L}{C}}$

$\tilde{V}_L = \tilde{I} \underline{Z}_L = \frac{V_0}{R} i\omega_0 L = i \frac{V_0}{R} \sqrt{\frac{L}{C}}$

if $\frac{R}{L} \ll \omega_0$
 then: $|V_L| = |V_C| \gg V_0$

$V_{out} = V_R = \frac{R}{R + \frac{1}{i\omega C} + i\omega L} V_{in}$
 $\left| \frac{V_R}{V_{in}} \right|^2 = \frac{\omega^2 (\Delta\omega)^2}{\omega^2 (\Delta\omega)^2 + (\omega^2 - \omega_0^2)^2}$
 $\Delta\omega = \frac{R}{L}$

